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# Random Forest Algorithm based Device Authentication in IoT

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***Abstract*—Distributed devices and data management systems are major components of IoT network. IoT security and privacy are of the utmost significance because there are already billions of IoT devices, applications, and services in use and more are going online every day. Authentication, confidentiality, integrity and attacks are some important security and privacy issues in IoT systems. Authentication is a crucial security issue in order to secure the Internet's devices and data. The process of identifying users or devices is known as authentication. In this paper, a non-linear machine learning algorithm is used for device authentication in the network. The proposed scheme adopts Belief-Desire-Intention (BDI) server agency with the Random Forest (RF) algorithm, using the context information of the IoT objects for authentication. The proposed authentication scheme operates as follows: Authentication request from IoT device to authentication server agency, generation of session key, upon request from an IoT device, validation of the IoT device using a username/password and session key. The authentication server calls the BDI model, which uses context based information of device to make authentication decisions using a random forest (classification and regression) algorithm. Simulation results show that the performance of the proposed intelligent authentication approach is better in terms of authentication accuracy, computational delay etc.**

***Index Terms*—IoT, Authentication, Machine Learning, Random Forest**

## I. Introduction

The Internet of Things (IoT) has become one of the most important technologies over the past several decades. IoT devices can provide computational systems with countless advantages. Modern IoT devices have actuators and smart chipsets, but in addition to hardware, they also feature application based software. IoT devices have the capacity to gather, exchange, and transmit data over the Internet without human intervention. IoT aims to develop self-aware, intelligent, and autonomous devices for infotainment, smart cities, and healthcare, among other applications [1, 2].

Despite the IoT's technical, social, and enormous economic potential, the development and deployment of various IoT infrastructure, services, and applications has been significantly hampered by security and privacy concerns [3, 4]. It's also crucial to understand how IoT enabled scenarios may present new design issues and requirements over traditional security and privacy schemes like identity access, security management, intrusion detection, and cryptography [5]–[7].

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IoT security and privacy are the backbone of the Internet. To exchange data, internet-connected IoT devices are linked to one another. Thus, if one of these devices is compromised, all of them are susceptible to compromise. IoT privacy protects sensitive information about people and organizations, whereas IoT security detects and prevents unauthorised access. Different techniques, including encryption, authentication, and permission, are used to meet the demand for IoT security [8].

Computational Intelligence (CI) principles have been successfully applied in modern contexts to discuss a variety of IoT issues, including scalability of the IoT network, aggregation & dissemination of information, routing, security & privacy issues like confidentiality, integrity, and authentication. CI offers adaptable algorithms that demonstrate intelligent IoT behaviour in dynamic and challenging situations. In response to diverse IoT difficulties, CI offers robustness, autonomy, and adaptability [9].

### *A. Problem Statement*

Username/Password schemes are no longer a reliable option for online authentication in IoT networks. Devices with different computational/communication technologies, hardware types, and embedded with different operating systems adopt traditional authentication schemes like cryptographic algorithms, signature based algorithms, and multi factor algorithms, which are complex and time consuming. In this context, with incomplete knowledge of the IoT network and device behaviour, implementing an intelligent authentication scheme in an efficient and adaptive way in a real time application is a challenging task. Hence, effective authentication of objects in IoT scenarios should be addressed carefully.

### *B. Our Contributions*

The proposed authentication scheme and authentication server cognitive agency are more practical and lightweight because the environmental data is only delivered when the user makes an authentication request. The proposed model works under different phases such as: 1) An authentication request from the IoT device, 2) If the response from the authentication server agency is positive, collect IoT device context information, 3) The authentication server cognitive agency executes the BDI model with the help of the Random Forest machine learning algorithm, 4) After analysing the device context information, if the response is positive, the device is authenticated.

The rest of the paper is organised as follows. Related work discussed in section 2. Section 3 presents the proposed RF-BDI based device authentication scheme. Section 4 discusses simulation and result analysis. Conclusions and future directions are discussed in Section 5.

## II. Related Work

Random Forest (RF) machine learning algorithm easily handles binary features, numerical features, and categorical features. RF offers a significant performance gain over a single tree and offers a lower error rate, excellent precision, and efficiency, along with exceptional noise resistance [10]. The RF algorithm is free of overfitting issues and local minimums. Since the decision trees are independent of one another, the likelihood that a local minimum problem or an overfitting problem will arise in the same location for various decision trees reduces exponentially, almost to zero, as the number of decision trees rises.

Cognitive Agents (CA) includes mental states, traits, and attitudes like belief, desire, and intention. The BDI paradigm empowers agents with the cognitive capacities necessary to deal with dynamic, uncertain, and complex situations [11]. To do specific tasks without interfering with host activity, BDI agents employ their own knowledge bases. The BDI model has three important attributes : an informational state about the world/network (beliefs), a motivational state to achieve the goals (desires), and a sequence of plans for action to be taken (intentions). A cognitive agency is used by a BDI to carry out duties on behalf of the user while functioning as an assistant to the user [12, 13].

Machine learning based trust aggregation authentication protocol is dicussed in [14]. The internet gateways for each device determine the total trust value based on the trust values of the data and behavior. Gateways omit a node from the authentication process if either the trust value is below a certain threshold or the authentication token is incorrect. The support vector machine technique used to calculated trust threshold values.

Author designed and enhanced 2-stage verification model based on ECC and biometric data [15]. The scheme implements specific security measures and protocol-designed logic that has been validated using the AVISPA tool. In order to authenticate and carry out the key agreement, the author created Modified Honey Encryption utilizing the Inverse Sampling-Conditional Probability Model Transform (MHE-IS-CPMT) protocol [16]. The MHE secures the user and device data with keys created by the ECC technique. By giving them plausible data instead of the genuine, sensitive smart home data, the proposed protocol confuses the system's enemies.

For Internet of Things (IoT) devices, a Process-based Pattern Authentication (PPA) system [17] that does not require a server to keep track of the login user's static password. The R-code and the symbol that users enter during registration are stored on the server, but the P-code—the actual password—changes with each time a user tries to log in. In this approach, users can create a pattern based on calculations from the P-code and R-code in the PPA pattern, and they can utilize an Artificial Neural Network (ANN) to authenticate themselves using their touch-based dynamic behaviors.

The IoT Devices registration process is performed through private blockchain. In order to cut the cost of calculation, the author mentioned the mutual authentication process, which is carried out without the assistance of RAC/Gate-Way-Node (GWN). The proposed protocol includes extra features including location-based authentication, a revocation phase based on blockchain technology, registration of IoTDs, and IoTD anonymity at the device level.

An effective Zero Knowledge-based authentication method [19] authenticates network devices without learning the identity of the user or disclosing any other data entered by the user. Developed the ZKNimble data encryption technique, which is mostly appropriate for portable devices. ZKNimble can be utilized for lawful users' encryption and decryption procedures after the user has been authenticated via Zero Knowledge Proof.

In order to handle Internet of Things (IoT) devices and provide security without taking into account a reliable third party, the author suggests an authentication technique [20] employing fog nodes. Fog node-based authentication provides trustworthy verification between data owners and the requester without relying on outside users. The storage system's single point of failure was successfully addressed by the fog nodes, which also provide numerous advantages by boosting throughput and cutting costs.

## III. Proposed Work

### A. Network Architecture

The general IoT network architecture for device authentication is as shown in figure 1. A typical IoT network is made up of remote, resource-constrained data nodes coupled to static server nodes that use routers to send data to the cloud. A network gateway transports the data from the routers to the cloud. Simple sensors to complicated systems with processors, memory, and communication are all examples of IoT devices.

### B. Random Forest Model

Proposed machine learning model for IoT device authentication is as shown in figure 2. Cognitive agency model comprises the Beliefs-Desires-Intention (BDI) model. In the IoT, as mobile devices use pervasive computing settings, context-aware authentication solutions are gaining popularity. They can be used in a variety of ways and enable safe authentication by studying the user's device behaviour. When used in conjunction with password-based authentication techniques, context-aware authentication systems add an extra degree of security. In the proposed research work, we offer a simple to implement BDI based cognitive agent driven context-aware authentication system for IoT devices. Cognitive agency executes Random Forest Machine Learning algorithm to authenticate/not authenticate IoT devices.

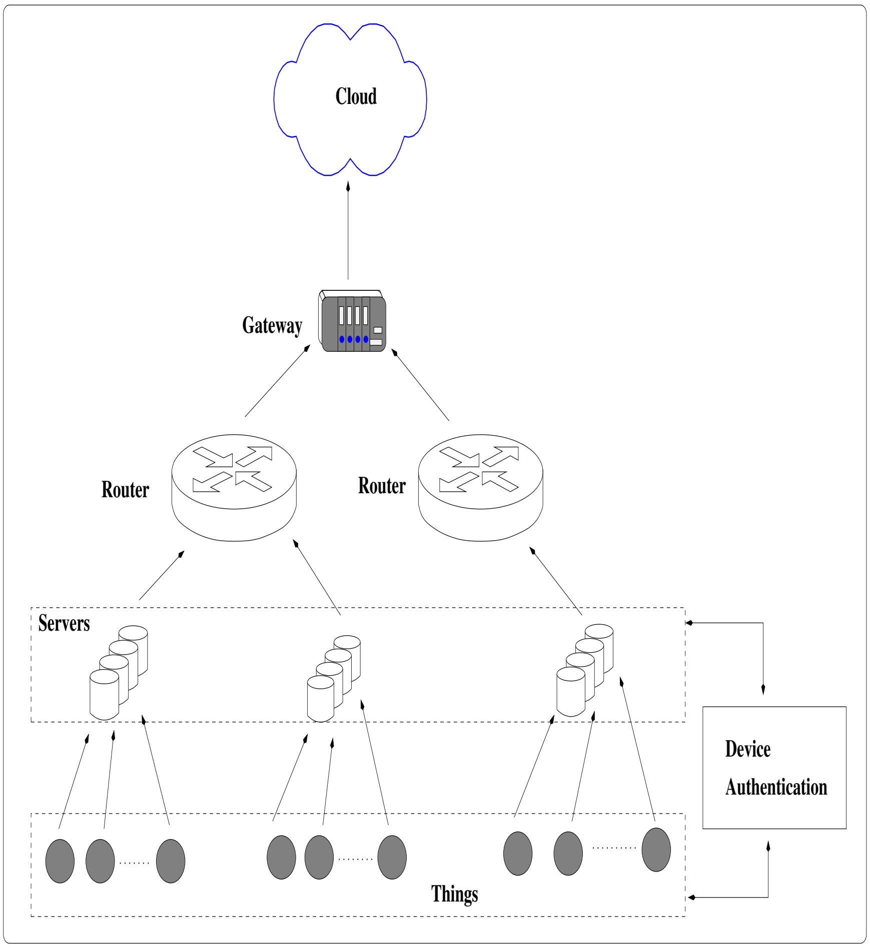


Fig. 1. Proposed IoT Architecture

*1) Initial Set-up Phase:* Authentication server when wants to know exactly who is accessing their information or site, it uses authentication. When an IoT device wants to know that the server is the system it purports to be, authentication is utilised. The IoT device must confirm its identity to the authentication server during authentication. In the proposed context aware device authentication system as an initial set-up phase, IoT devices request for session. In turn, the authentication server grants the session by generating a session key $(S_k)$.

Once the user name and password authentication is successful, the authentication server asks a question i.e., challenge and the IoT device must respond with a correct answer i.e., the same session key $(S_k)$ already provided by the authentication server, in order to be authenticated. Once the authentication is successful, cognitive agency of the authentication server collects the information of device context. This authentication process simply establishes and confirms the identity of the IoT device.

*2) Collection of Device Context Information:* IoT device context defines a set of events in a specific entity, location, and time period situation. In the suggested context-aware authentication system, we make an effort to employ resources often present in IoT devices to monitor user behavior in various scenarios and events when the user is engaged in the ubiquitous area. In our proposed system, we consider IoT device context information such as device identity $(D_{id})$, requested activity $(D_{act})$, GPS position of the device $(D_{loc})$, time zone $(D_{time})$, and device characteristics $(D_{char})$ like running applications, device hardware type and device OS, etc.

*3) Authentication Server Cognitive Agency:* An authentication server oversees the processes that allow users to gain access to a network, application, or system. Before connecting to a server, devices demonstrate that they are who they claim to be. This delicate task is handled by an authentication server. Authentication servers can be embedded in switches, dedicated computers, or network servers. Authentication servers play an important role in keeping critical assets safe and secure. In the proposed scheme, the authentication server (AS) performs the following tasks: 1) Generation of session key, upon request from an IoT device, 2) Validation of IoT device using username/password and session key, 3) Gathering of context based information of devices to be authenticated, 4) The authentication server calls the BDI model, which uses context based information of device to make an authentication

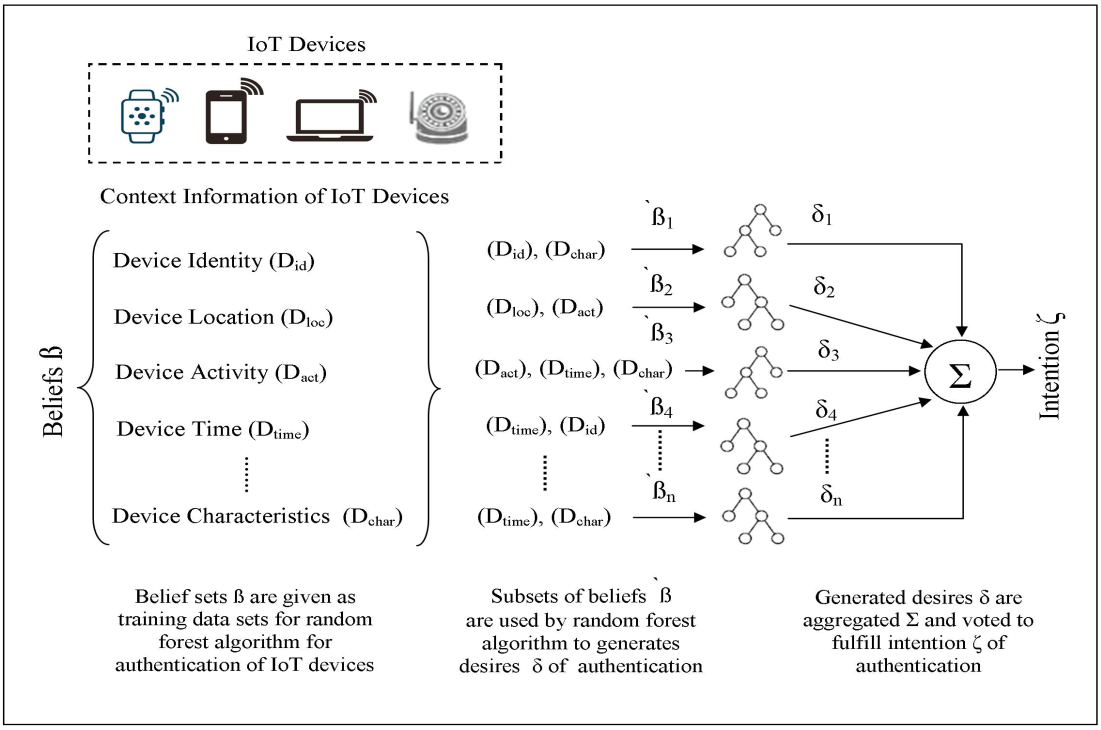


Fig. 2. Proposed Random Forest Algorithm

decision using a random forest (classification and regression) algorithm.

*4) Final Authentication Phase:* Context aware authentication is provided by authentication server agency by executing BDI model shown in fig. 2:

- Belief Generations: Context aware information of IoT devices $D_n$ is acquired after validation process. This context based information is nothing but set of beliefs $\beta$ where $(\beta = D_{id}, D_{loc}, D_{act}, D_{time}, D_{char})$ and $\grave{\beta}$ is the subsets of beliefs used as training data set in random forest algorithm where $(\grave{\beta}_1 = D_{id}, D_{char})$, $(\grave{\beta}_2 = D_{loc}, D_{act})$ and so on. The process of generating beliefs subset from sample set of beliefs is called as bagging. The table III shows sample belief sets for several IoT devices with different characteristics & capabilities. Beliefs are static in nature that do not change their values or behaviour during execution. The belief analyser divides beliefs into groups of previous and present beliefs. The historical information of an IoT device that has previously been verified with a server is contained in the past beliefs, whilst the data related to the current context environment is kept in the current belief sets. Further these updated belief sets (attributes) and subsets (instances) are considered for desire generation $\delta$.
- Desire Generations: In general beliefs $\beta$ lead to generation of desires $\delta$. AS uses non linear regression based random forest with kernel functions to generate desires $(\delta_1, \delta_2, ..., \delta_n)$ by sub sets of beliefs $(\grave{\beta}_1, \grave{\beta}_2, ..., \grave{\beta}_n)$ (as discussed in section C computational models).
We used information gain, or the Gini index, to choose instances for our proposed model from the data set. The beliefs are selected for the training phase based on the Gini index values for all potential attributes and the instances of IoT devices for each of those qualities as indicated in eqn. (1) [14].

$$Gini\ Index = 1 - \sum_{\grave{\beta}} P_{\grave{\beta}}^2 \tag{1}$$

Where, $P_{\grave{\beta}}$ is the probability of each class of context information of IoT devices.
- Intention Generations: The core part of the server is to generate intentions $\zeta$. A determined desires may be either to authenticate the device or not authenticate. All desires are aggregated and final intention of authentication with server is declared based on voting mechanisms.
In the proposed scheme revision of desires is required in two conditions. First, if any cell in the data set is empty or its value is zero, the revision model is needed and KeRf scheme is used for this purpose to generate more precise desires (as shown in section C computational model). Second, if the intentions aggregated are equal, the voting process is unable to decide the intention of the cognitive agency. After the revision and final voting process is finished, the possible outcome is either to authenticate or not authenticate, which is communicated to the IoT device by server. If the final call is to be authenticated, then the server stores the context information of that device in its

database for future authentication.

## IV. Simulation

The PyCharm IDE is a process based discrete event simulator used for machine learning algorithms. In this section, the simulation model, simulation inputs and result analysis are discussed.

### *A. Simulation Inputs*

To illustrate the results of the proposed scheme, the simulation input parameters are as follows: number of IoT devices $D_n$ = 20 - 100, communication range of each IoT device $C_r$ = 50 - 100m, area of the IoT network $a = 500 * 500m^2$, context information size M = 512 bits to 4096 bits.

### *B. Result Analysis*

To test the operative effectiveness of the proposed scheme i.e., cognitive agent based random forest learning algorithm approach for IoT device authentication. We have analysed some of the performance metrics which are mentioned in the underneath graphs. Simulation results show that the proposed algorithm is effective and efficient in terms of delay management and authentication accuracy.

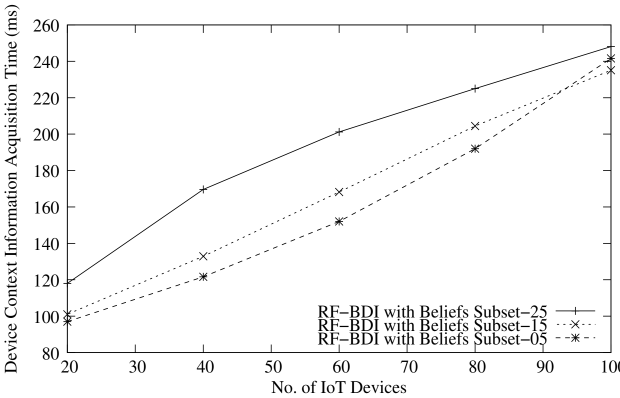


Fig. 3. Device Context Information Acquisition Time Vs. No. of IoT Devices

Time taken by the authentication server agency to collect the each IoT devices context information is shown in fig. 3. The context information acquisition delay relatively increases with increase in the number of IoT devices in the network. The delay increases with in increases in device beliefs and subsets of beliefs.

Authentication intentions is the measure of the RF-BDI model accuracy i.e., how many times the machine learning algorithm and cognitive agency has correctly predicted the authentication using context information. From graph 4 we observe that as number of beliefs set context information increases the accuracy level also increases. Authentication accuracy of the proposed scheme is higher than the traditional algorithms.

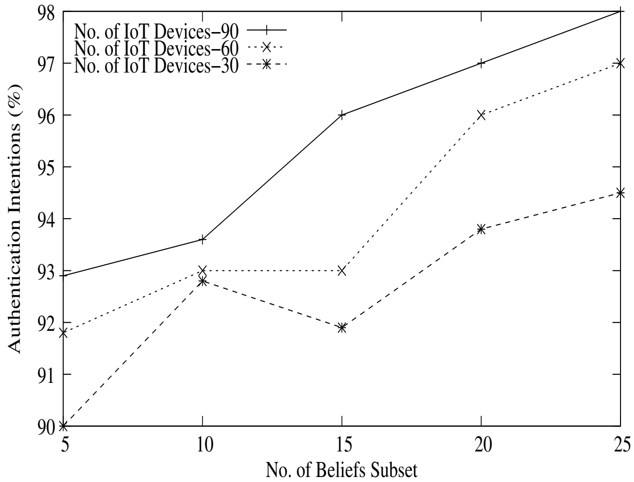


Fig. 4. Authentication Intentions Vs. No. of Beliefs Subset

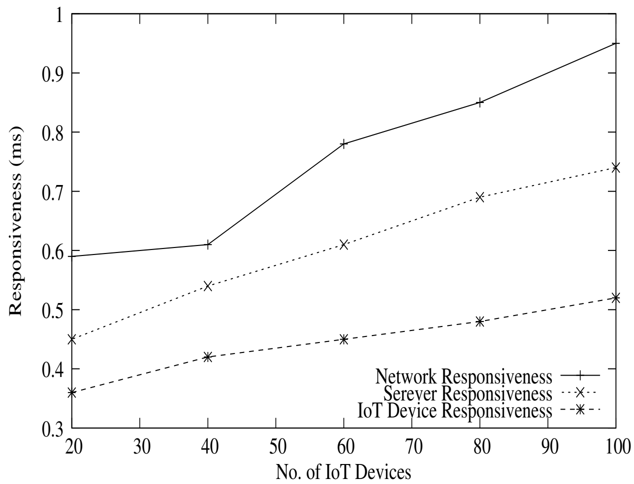


Fig. 5. Responsiveness Vs. No. of IoT Devices

From fig. 5 illustrates the response times for the IoT deviec, server and IoT network for varing IoT devices during authentication attempts. We observe that the total time of network authentication attempt does not exceed one second.

The proposed RF-BDI model for device authentication in IOT network yields accuracy upto (98.1%), with precision rate around (97.2%), the sensitivity and recall of the RF-BDI model is (96%) & (96.59%) respectively with very low false positive rate of (3.3%). The obtained results show that, in complex unprotected networks considering context information of IoT devices, device authentication accuracy, computational latency are better.

## V. Conclusion

In the proposed intelligent authentication scheme, we have integrated Belief-Desire-Intention agents with a random forest machine learning algorithm to make authentication decisions using the context information of the IoT devices. The context information of IoT devices, i.e., the belief set, is explicitly used to generate desires of intentions with the server, and finally, a precise and secure intention is generated through proper voting and secure authentication between the IoT device and server. The generation of beliefs, desires and intentions in the RF-BDI model is determined using a non-linear regression scheme. Server uses device context information to build a user model in order to make an authentication decision over an authentication request. Simulation results show that authentication accuracy and computational delay are better when compared with existing authentication algorithms. Authentication is performed between the server and IoT devices in the proposed scheme; further it can also be achieved at the router, gateway, edge, and cloud levels.

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